

Introduction

- In Mandarin, one type of Alternative Questions (AltQs) is formed with a specific morpheme *háishì*.

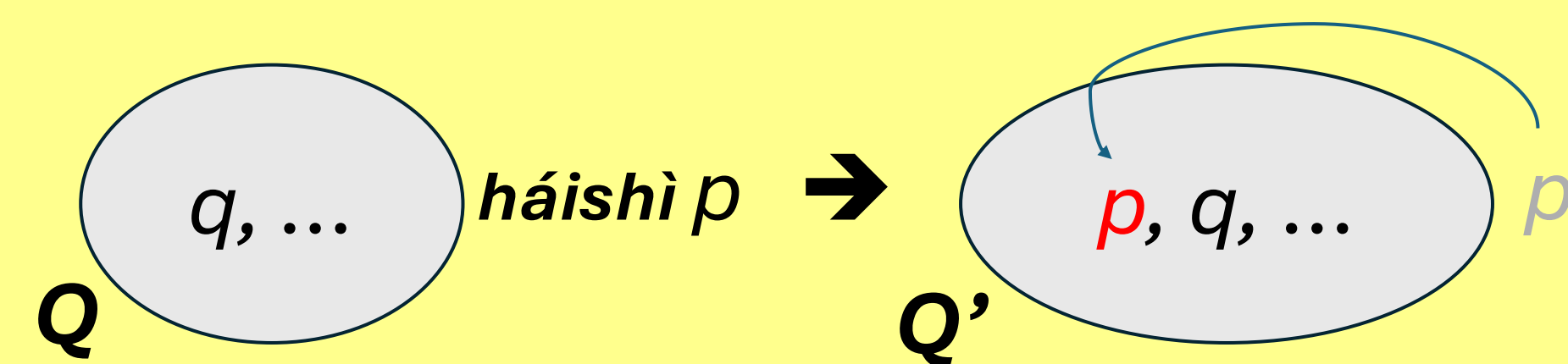
(1) Ā Máo xǐhuān hē kāfēi (ne,) { *huòzhě/ háishì } xǐhuān hē chá (ne)?
A Mao like drink coffee NE or or like drink tea NE
'Which one is true: A Mao likes drinking coffee or tea?'

- Previous analyses all assume that *háishì* relates objects of the same sort.
- Semantic type-wise, it means *háishì* connects objects of the same type: propositions, questions, etc. (Li & Thompson 1981, Huang 1991, Erlewine 2025, a.o.)

- I call these analyses **symmetric** and argue **against such a symmetry**.

- Key:** [S_L *háishì* S_R] expresses an **asymmetric** mapping from a proposition p and a question Q to form a new question Q' , roughly “ Q or p ?”.

(2) [S_L *háishì* S_R]



- The proposition, p , is added to the set of possible answers for Q .
- If Q is a Polar Q, then its positive answer q forms an AltQ with p : “ q or p ?”

- I will present **two empirical asymmetries** in *háishì*-constructions involving **question particles** and **wh-question disjunctions**.

Challenge 1: PolQ particle

Observe the distribution of the **polar question particle** *ma*:

- Ma* occurs only with matrix PolQ (not other kinds of questions).

(7) Ā Máo xǐhuān hē kāfēi **ma**? (8) Ā Máo xǐhuān hē shénme **ma**?
A Mao like drink coffee QP A Mao like drink what QP
(I) ‘Does A Mao like drinking coffee?’ # ‘What does A Mao like drinking?’
(D) # ‘A Mao likes drinking coffee.’

→ It is thus commonly understood as marking a PolQ (Cheng 1991, Paul 2014, Pan 2019, a.o.)

- In AltQs, the right disjunct (S_R) of *háishì* resists *ma*-marked.

- When S_R is not *ma*-marked, S_L is compatible with *ma* (and *ne*, cf. ex. (1)).

(9) Ā Máo xǐhuān hē kāfēi **ma**, háishì xǐhuān hē chá (ne)? ✓ [... *háishì* ... (ne)]
A Mao like drink coffee QP or like drink tea NE
'Does A Mao like drinking coffee, or does he like drinking tea?'

- When S_R is *ma*-marked, the sentence is ungrammatical.

(10) *Ā Máo xǐhuān hē kāfēi { **ma**/ ne/ ∅ }, háishì xǐhuān hē chá **ma**? * [... *háishì* ... *ma*]
A Mao like drink coffee QP NE or like drink tea QP
(Judgment from Mainland Mandarin speakers; cf. Taiwan Mandarin, Hsieh 2026)

Assuming that *ma* tracks the type of S , I suggest the anomaly can be explained if S_R cannot be a PolQ. What it is in (9) is a proposition.

Challenge 2: wh-question

Observe the acceptability of *wh*-questions:

- The right disjunct (S_R) of *háishì* cannot be a *wh*-question.

(11) Nǐ de diànhuà shì duōshǎo, { huòzhě /*háishì } nǐ de yóuxiāng shì shénme? [*whQ*+*whQ*]
2sg DE phone number be how many or or you DE email be what
'(To recover the username, please tell me) what is your phone number, or what is your email?'
(12) Nǐ dǎsuàn zài nǎr zū chē (ne), háishì dǎsuàn jiè yí-liàng (ne)? [*whQ* + *háishì*(p)]
2sg plan LOC where rent car NE or plan borrow one-CL NE
'Where do you plan to rent a car, or do you plan to borrow one?' (after Ciardelli et al., 2018)

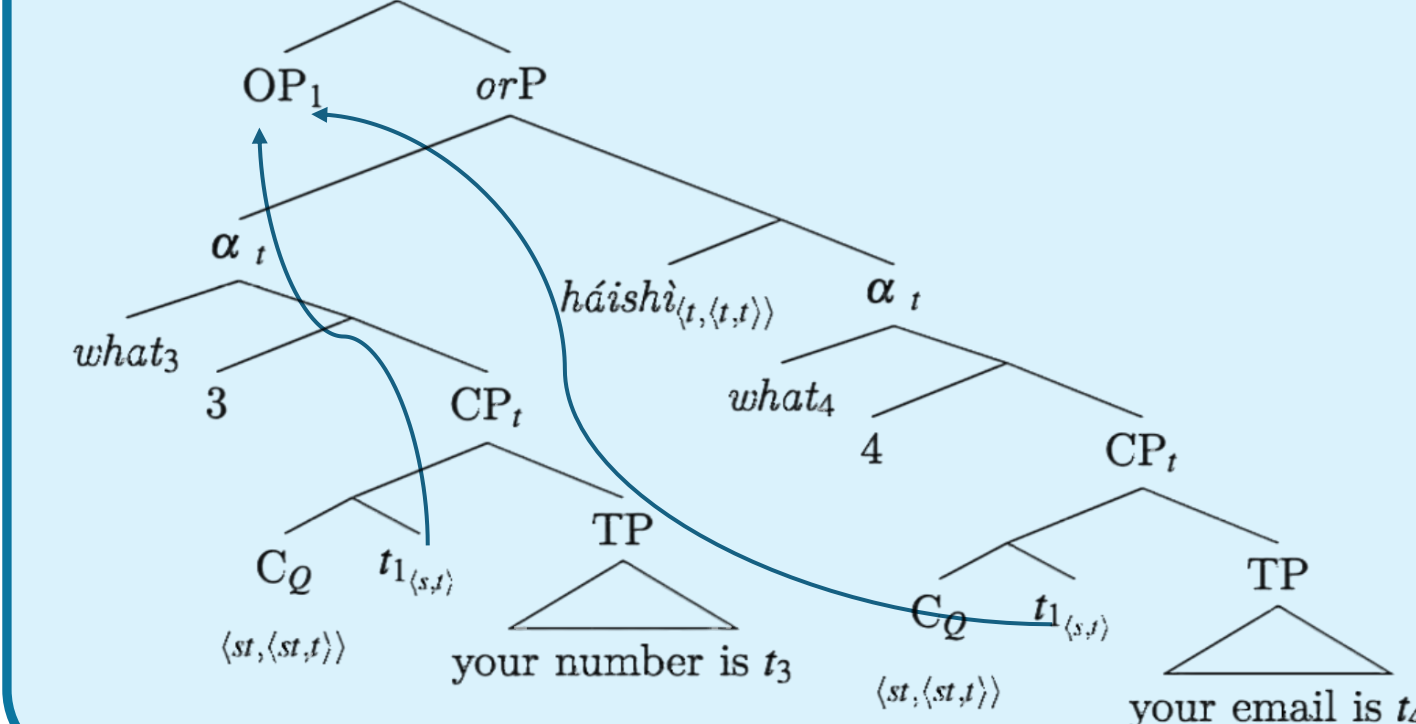
- Using a sequence of two *wh*-questions is bad.
- But when the right clause is arguably not a question, the sequence is acceptable.

(H3) If arguments of *háishì* can be question types (Erlewine 2025):

- E.g., $\llbracket p \rrbracket^{alt} = \{p\}$ → Both disjuncts are of the type $\langle st, t \rangle$,
 $\llbracket Q \rrbracket^{alt} = \{p_1, p_2, \dots\}$ no semantic explanation for (10) and (11).

(H2) If *háishì* is a logical connective (e.g., Huang 1991), we might overgenerate the following **plausible but unattested** LF (vF&H 2021):

*LF: [OP_1 [*what*₃ [[C_Q t_1] your-number-is t_3] *háishì* _{$\langle t, \langle t, t \rangle \rangle$} [*what*₄ [[C_Q t_1] your-email-is t_4]]]]



→ Again, if S_R cannot be a question, (11) is predicted.

Background and previous analyses

In (1), *háishì* connects two seemingly declarative clauses S_L and S_R .

(1) Ā Máo xǐhuān hē kāfēi (ne,) { *huòzhě/ háishì } xǐhuān hē chá (ne)?
A Mao like drink coffee NE or or like drink tea NE
'Which one is true: A Mao likes drinking coffee or tea?'

→ Neither clause is marked by any overt interrogative morpheme (*ne* is not one). But the disjuncts might as well be a question:

Ā Máo xǐhuān hē kāfēi is ambiguous b/n declarative and interrogative.

- Assuming symmetry between two clauses, we have (at least) three possibilities for what type they denote: **propositions**, **questions**, or objects with flexible but matching types.

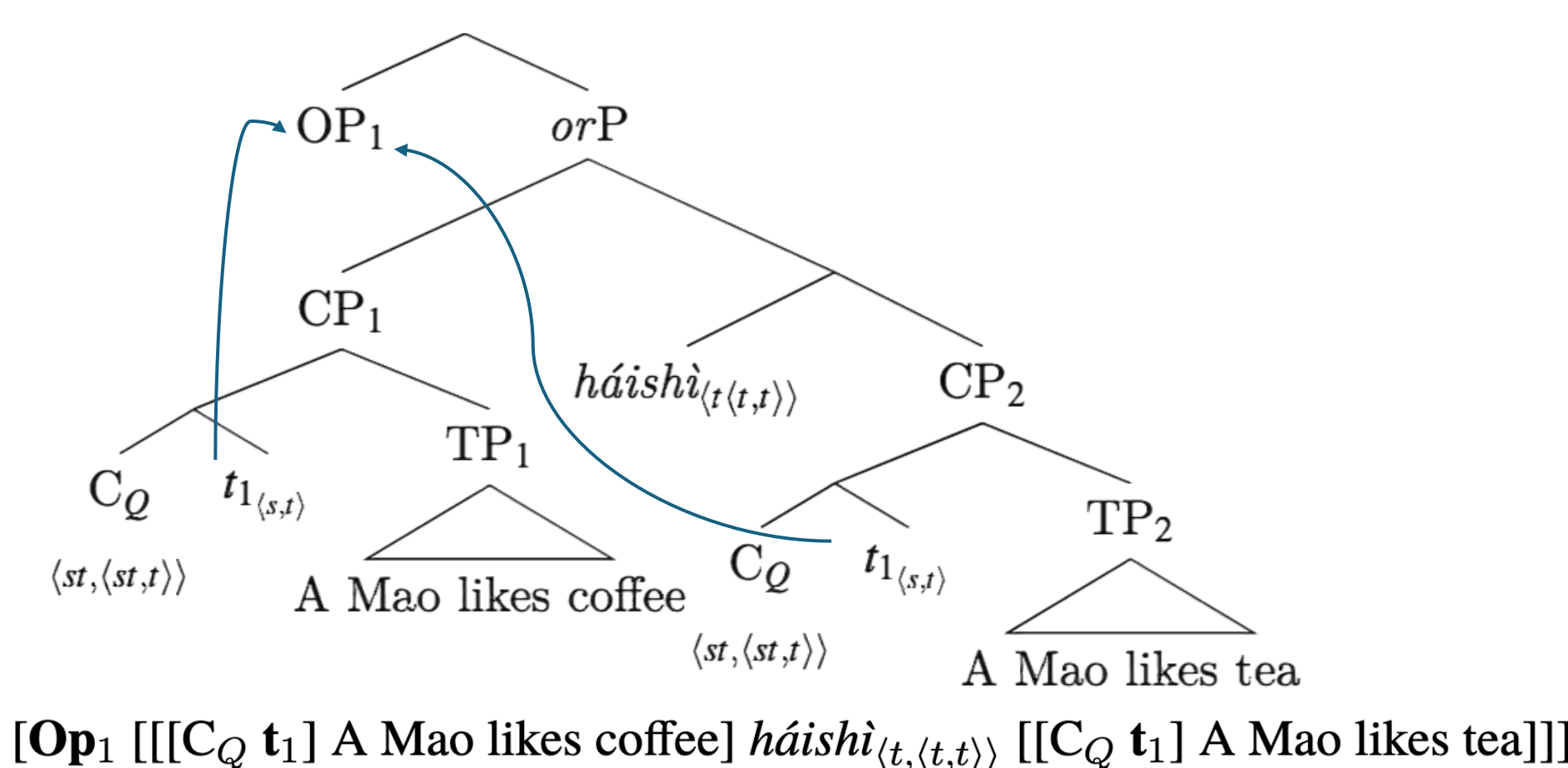
- Three existing hypotheses discussed. **Semantically, all these analyses assume that the disjuncts connected by *háishì* are symmetric.**

- H1 (refuted by recursion fact): S_L and S_R denotes propositions, and the combination expresses an AltQ with those two propositions as its possible answers (Li & Thompson 1981).

(3) $\llbracket háishì \rrbracket = \lambda p_{st} \cdot \lambda q_{st} \cdot \lambda r_{st} \cdot r = p \vee r = q$.

- H2: S_L and S_R denotes propositions, and *háishì* is a **logical disjunction**, the AltQ derived via scoping the disjunction over an interrogative C (Huang 1991, after Karttunen 1977; von Stechow & Heim 2021).

(4) $\llbracket háishì \rrbracket = \lambda p_t \cdot \lambda q_t \cdot p \vee q$.



- H3: *háishì* expresses **set union** $\cup$ in the domain of Focus Semantic Value (Erlewine 2025, after Rooth 1992, Beck 2006). The denotation of *háishì* is compatible with disjuncts of varying semantic types $\langle st, t \rangle$, $\langle s, t \rangle$, e , etc.).

(5) $\llbracket \beta háishì \alpha \rrbracket^{alt} = \llbracket \beta \rrbracket^{alt} \cup \llbracket \alpha \rrbracket^{alt}$

A specific instance of *háishì* would look like (6), where α and β are of question type.

(6) $\llbracket háishì \rrbracket = \lambda Q_{stt} \cdot \lambda Q'_{stt} \cdot Q \cup Q'$.

Proposal and discussions

Proposal: The observed semantic asymmetries are reflected in the semantics of *háishì*.

(14) $\llbracket háishì \rrbracket^w := \lambda p_{st} \cdot \lambda Q_{stt} \cdot \lambda r_{st} \cdot [Q(r) \vee r = p]$

→ S_L must be a question, and S_R must be a proposition..

→ When S_L is a PolQ, a monopolar analysis is assumed: Q denotes a singleton set $\{p\}$. (Roberts 1996, Uegaki 2014).

- One might wonder if *háishì* does not scope over the preceding questions in (8) and (12):

Q1: [S_L]? Q2: [*Háishì* S_R]?

→ To maintain the semantics as in (14), one can postulate a pronoun anaphoric to the salient preceding question.

Q1: [S_L]? Q2: [*pro*_{Q1} *Háishì* S_R]?

Consequences of the semantics in (14):

- I hypothesize that clause-medial *háishì* AltQs in Mandarin, such as “Does A Mao like coffee or tea?”, undergoes the same composition as the biclausal one in (1), plus ellipsis.
- For declarative use of *háishì*, lexical ambiguity is assumed (cf. Erlewine’s uniformed analysis (2025)).
- Another disjunct *huòzhě* **never** forms AltQs in Mandarin (1), its semantics also needing restriction to block these constructions altogether.